

ch 4.7.5

rotation: 2 sec

clock rate: 100 MHz

$$2 \text{ sec} = \frac{1}{100 \text{ M}} \times n \Rightarrow n = 200 \text{ M}$$

$$\text{ceil}(\log 200 \text{ M}) = 28$$

we need 18 bits in Mod m counter

parameter in Mod_m counter:

$$N = 18$$

$$M = 200000$$

In this chapter, designs are without ASM.

Inputs { en
dir
tick from Mod_m
input message with B bits width

n_reg, n_next $\Rightarrow$ { reset | n_reg = 0
out_reg = input[B-1:B-4]
operation | n_next = n_reg + 1

$out_next \neq out_reg < 1$ or $out_reg > 1$
~~if dir = 0~~
if dir = 1

if $n_reg < B$
operation

else $n_st.p$
 $out_next = out_reg$

$out_ptr = out_ptr - reg[B-1, B-4]$

case $[en, dir, tick]$

$\rightarrow 0, X, X$ or $X, X, 0$

$out_next = out_reg$

$n_next = n_reg$

$\rightarrow 1, 1, 1$

$\rightarrow 1, 0, 1$

+ verification
circuit
+ test circuit

0123456789

↓

0123

↓ left

1234

↓

0123

↓

9012

right